

Demand and Welfare

Session 4

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Introduction

Reading

Deaton is in your `reading/` folder on the hub, or PDF.

- Deaton, chs. 4 (§4.2 on Engel curves) and 5 (all)
- Deaton & Muellbauer (1980), "An Almost Ideal Demand System," *AER* 70:312–326
- Lewbel (1991), "The rank of demand systems," *Econometrica* 59:711–730
- Ligon, "Estimating Household Welfare from Disaggregate Expenditures" (CUDARE working paper, 2020) — the CFE system we estimate today. In your `reading/` folder on the hub, or eScholarship.

Engel's Laws and Curves

Engel may be the first to use household data in statistical analysis, and used household budget data from Belgium and Saxony to propound what is sometimes called "Engel's (first) Law":

"The greater the income, the smaller the relative percentage of outlay for subsistence."

Budget shares and Engel curves

Let $s_{ij} = p_j q_{ij} / x_i$ be household i 's budget share on good j , x_i total expenditure. The Working–Leser form,

$$s_{ij} = \alpha_j + \beta_j \log x_i + \gamma_j' z_i + u_{ij},$$

fits food shares remarkably well across countries and decades.

- Homothetic utility implies $\beta_{\text{food}} = 0$.
- $\beta_{\text{food}} < 0$ is Engel's law
- z_i : household size and composition, region, season
- Estimated by OLS, clustered on the EA

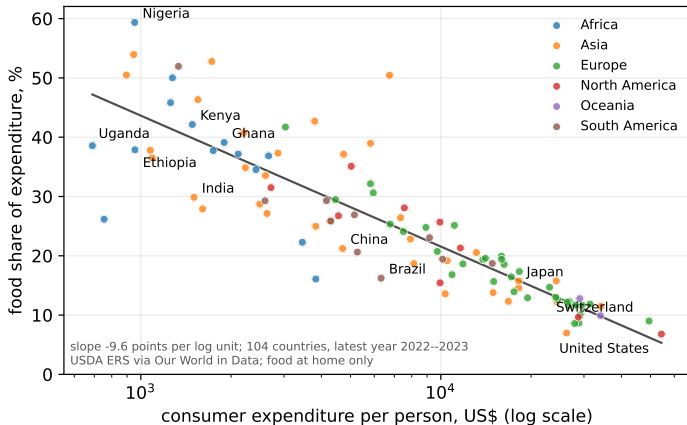
Engel's Second Law

A sort of corollary to the first law, sometimes translated as
"The proportion of the outgo used for food, other things being equal, is the best measure of the material standard of living of a population."

As a welfare measure, this solves some problems that the consumption aggregate has. In particular:

- Is a dimensionless share, so don't need to worry at all about price level, etc.
- If household size/scale can be expressed using Adult Equivalents A_i^θ , then those cancel also.
- Appears to be naturally comparable across time or place.

Engel's law across countries



One point per country, latest year.

Other Important Elements of Demand Estimation

Equivalence scales from behaviour

Session 3 *assumed* a scale A_i^θ . Demand analysis lets you estimate one.

- **Engel method:** two households are equally well off if they have the same food share. Scale = the expenditure ratio that equalizes it.
- **Rothbarth method:** use expenditure on adult goods (alcohol, tobacco, adult clothing) as the indicator instead.
- The two disagree, systematically and by a lot. Deaton ch. 4 explains why: the Engel method conflates the cost of a child with the child's effect on the household's needs for food specifically.

Prices, and where to get them (or not)

LSMS surveys may not carry reliable data on prices. They carry unit values, and unit values are contaminated by quality choice:

$$\log v_{ij} = \log p_j + \underbrace{\psi_j \log x_i}_{\text{quality}} + \epsilon_{ij}.$$

Use the market comparison

Deaton's solution exploits the design: *within a market*, all households face the same price. So

- regress log unit prices $\log v_{ij}$ on $\log x_i$ and household characteristics *with market fixed effects* to strip out quality;
- the cluster effects are the price variation you wanted.

This is the single best argument for caring about v in the index.

→ notebook

The rank of a demand system

Stack the Engel curves. The *rank* of a demand system (Lewbel 1991) is the dimension of the space spanned by the budget shares viewed as functions of total expenditure.

- rank 1: shares do not vary with x — homothetic
- rank 2: Working–Leser, AIDS
- rank 3: QUAIDS; Gorman's bound for exactly aggregable systems
- rank n : Constant Frisch Elasticity (CFE) (coming up)

Why you should care: **if the rank exceeds one, no single price index can deflate nominal expenditure so as to keep it monotone in utility when relative prices move.** Rank r needs r indices.

Constant Frisch Elasticity Demands

Constant Frisch Elasticity demands

Let λ_i be household i 's marginal utility of expenditure.
Frischian (λ -constant) demands take the CFE form

$$f_j(p, \lambda) = \left(\frac{\alpha_j}{\lambda p_j} \right)^{\beta_j}, \quad U(c) = \sum_j \alpha_j \beta_j \frac{c_j^{1-1/\beta_j} - 1}{\beta_j - 1}.$$

- β_j is the **Frisch elasticity** of demand for good j :
necessities small, luxuries large
- $\beta_j = \beta$ for all j gives CES; $\beta \rightarrow 1$ gives Cobb–Douglas
- unrestricted rank!

The estimating equation

Take logs of expenditure on good j . Writing $w_i \equiv -\log \lambda_i$,

$$\log x_{it}^j = a_t^j + \beta_j w_{it} + \gamma_j' d_{it} + \varepsilon_{it}^j.$$

This is a **one-factor model**. The $(it) \times j$ matrix of log expenditures, once good-time effects a_t^j and characteristics d_{it} are swept out, is rank one: loadings β , factor w .

- estimate by SVD of the residual matrix, with missing data
- d_{it} enters as good-specific *Barten scales*, not "independence of base" adult equivalents.
- needs **no prices** and **no total expenditure** — only expenditures on a subset of goods

$w = -\log \lambda$ as a welfare measure

The marginal utility of expenditure λ falls as a household gets better off. Hence $w = -\log \lambda$ increases with welfare.

- Comparable across households facing different prices — the prices are in a_t^j , not in w
- Comparable across household compositions — composition is in $\gamma_j' d$
- Derived from behaviour, not from an assumed equivalence scale or an assumed price index

Price Changes

Welfare consequences of a price change

How much money makes household i as well off at prices p^1 as it was at p^0 ? With an expenditure function $e(u, p)$,

$$CV_i = e(u_i^0, p^1) - e(u_i^0, p^0), \quad u_i^0 = v(x_i, p^0).$$

A first-order welfare approximation

For a small change in the price of one good j , compensating variation is approximately

$$CV_i \approx q_{ij} \Delta p_j = s_{ij} x_i \frac{\Delta p_j}{p_j}.$$

- to first order you need only the *budget share* — no elasticities
- the distributional incidence follows from how s_{ij} varies with x_i , which is the Engel curve you already estimated
- the exact figure needs the demand system, and says how far the approximation can be trusted

Exercises

Exercise 1: compare Engel curves

The Working-Leser cell uses Section 9A alone in its non-food denominator.

1. Interact $\log x$ with `Rural`. Does the slope differ between urban and rural households?
2. Re-estimate on the 2012-13 wave, which has the same section. Has β moved in five years?

Exercise 1: compare expenditure coverage

Uganda's `nonfood_expenditures()` includes frequently and infrequently purchased non-food items. Put them on a common recall basis:

- The questionnaire specifies 30 days for non-durables and 365 for semi-durables in 2019-20 (see session 3).
- Use `uga.categorical_mapping['nonfood_items']` to identify items.
- Repeat the estimation for Uganda 2019-20. Is the share still flat over most of the distribution? How do the composition coefficients change?

Exercise 1: account for clustered sampling

The reported standard errors allow heteroskedasticity but don't account for dependence within an EA.

- Cluster on the EA, using d.v.
- By how much do the standard errors grow?
- Is that the $\sqrt{\text{deff}}$ of session 1?

Exercise 2: equivalence scales

Implement the Engel equivalence scale.

Find the expenditure ratio that equates the predicted food share of:

- a household with two adults and two children;
- a household with two adults.

Compare with the θ -scale from session 2.

Exercise 3: who bears the price change?

For the 2005-06 to 2009-10 price change:

1. Compute the first-order welfare cost at each decile of the 2005-06 food budget, using predicted budget shares.
2. Compute exact compensating variation at the same deciles.
3. Who loses most in shillings? Who loses most as a share of the budget? Are those the same households?

Exercise 4: define regional markets

Re-estimate the CFE system with the four regions as markets. Use `market='Region'` on both tables instead of assigning every household to one national market.

- How much do the β_j move?
- What does π look like by region?
- What did one national market assume away?

Watch your kernel's memory while you do it.

Exercise 5: compare welfare measures

Regress w on log total food expenditure.

- How closely do the two measures agree?
- For which households do they disagree?
- What adjustments does the demand-based measure make that expenditure alone doesn't?